



Universal Generalization

Let $x \in \mathbb{R}$

Demonstrate $P(x)$

Therefore $\forall x \in \mathbb{R}, P(x)$

Prove

$$\forall x, y \in \mathbb{R}, (-x)y = -(xy)$$

$$\text{Let } x, y \in \mathbb{R}$$

$$-x + x = 0$$

$$(-x + x)y = 0y$$

$$-xy + xy = 0 \quad \text{add } -(xy)$$

$$(-x)y + xy - xy = -(xy)$$

Therefore, $\forall x, y \in \mathbb{R} \quad (-x)y = -xy$

Prove by contradiction

To prove P

Assume $\neg P$

derive Q and $\neg Q$

Therefore P

Prove $0 < 1$

Assume $0 \nless 1$

$$0 = 1$$

$$1 < 0$$

$$1 < 0$$

$$1 + (-1) < 0 + (-1)$$

$$0 < -1$$

$$0 \cdot (-1) < (-1) \cdot (-1)$$

$$0 < 1$$

we have $0 < 1$ and $0 \neq 1$

Therefore $0 < 1$

Prove $1+1 \neq 1$

Assume $1+1 = 1$

$$1+1-1 = 1-1$$

$$1 = 0$$

but $1 \neq 0$ zero

Contradiction

Prove $1+1 \neq 0$

Assume $1+1 = 0$

we know $0 < 1$

$$1 < 1+1$$

but $0 < 1$

contradiction

Proof by Contradiction within Universal Generalization

$$\forall x \in \mathbb{R}, P(x)$$

$$\text{Let } x \in \mathbb{R}$$

$$\text{Assume } \neg P(x)$$

$$\text{derive } Q \text{ and } \neg Q$$

$$\text{Therefore } P(x)$$

$$\text{Therefore, } \forall x \in \mathbb{R}, P(x)$$

$$\forall x \in \mathbb{R}, x \neq -x$$

$$\text{let } x \in \mathbb{R}$$

$$\text{Assume } x = -x$$

$$x + x = 0$$

$$\frac{1}{x} \cdot x + \frac{1}{x} \cdot x = 0$$

$$1 + 1 = 0$$

$$1 + 1 \neq 0$$

contradiction

Therefore, $x \neq -x$

Direct Proof

If P then Q

Assume P

Show Q

Therefore If P then Q

$\forall x \in \mathbb{R}, \text{ if } x < 0 \text{ then } 0 < -x$

Let $x \in \mathbb{R}$

Assume $x < 0$ $0 < -x$

$$x - x < 0 - x$$

$$0 < -x$$

Therefore, if $x < 0$, $0 < -x$

$\forall x, y, a \in \mathbb{R}$, if $a + x = a + y$
then $x = y$

Let $x, y, a \in \mathbb{R}$

Assume $a + x = a + y$... $x = y$

$$a + x + (-a) = a + y + (-a)$$

$$x = y$$

Therefore, if $a + x = a + y$

$\forall x, y, z \in \mathbb{R}$, If $x < y$ and $z < 0$, then $yz < xz$

let $x, y, z \in \mathbb{R}$

Assume $x < y$ and $z < 0$
 $-z > 0$

$$-xz < -yz \quad | +xz|$$

$$-xz + xz < -yz + xz$$

$$0 < -yz + xz \quad | +yz)$$

$$yz < -yz + xz + yz$$

$$yz < \underline{xz}$$

Therefore

Combine Direct Proof with Proof by contra

Prove IF P then Q

Assume P

Assume $\neg Q$

derive R and $\neg R$

Therefore Q

Therefore IF P then Q

$\forall x \in \mathbb{R}, \text{ IF } 0 < x, \text{ then } 0 < x^{-1}$

Let $x \in \mathbb{R}$

Assume $0 < x$

Assume $0 \geq x^{-1}$

$0x \geq x^{-1} \cdot x$

$$0 \geq 1$$

false

$$\text{Therefore } 0 < x^{-1}$$

Therefore, if P then Q

Existential Generalization

$$\exists x \in \mathbb{R}, x < 0$$

$$\text{Let } x = -1$$

$$\text{Since } 0 < 1$$

$$0 - 1 < 1 - 1$$

$$-1 < 0$$

$$\text{Therefore } x < 0$$

To prove $\exists x \in \mathbb{R}, P(x)$

let $x = \boxed{}$ some val.

Demonstrate $P(x)$

therefore

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x < y$$

$$\text{Let } x \in \mathbb{R}$$

$$\text{Let } y = x + 1$$

$$\text{we have } 0 < 1$$

$$x + 0 < x + 1$$

$$x + 0 < y$$

$$\text{So, } x < y$$

$$\text{Therefore } \exists y \in \mathbb{R}, x < y$$

$$\text{Therefore, } \forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x < y$$

$$\forall x, y \in \mathbb{R}, \text{ If } x < y, \text{ then}$$

$$\exists z \in \mathbb{R}, x < z < y$$

$$\text{Let } x, y \in \mathbb{R}$$

$$\text{Assume } x < y$$

$$\text{let } z = \frac{x + y}{2}$$

$$\text{Since } x < y \rightarrow x+x < y+x$$

$$\frac{2x}{2} < \frac{y+x}{2}$$

$$x < \boxed{\frac{y+x}{2}} \rightarrow x < z$$

$$\text{Since } x < y \rightarrow x+y < 2y$$

$$\boxed{\frac{x+y}{2}} < y \rightarrow z < y$$

$$\text{Therefore, } x < z < y$$

Prove Implication by contraposition

If P then Q

Assume $\neg Q$

demonstrate $\neg P$

Therefore If P then Q

$\forall x, y \in \mathbb{R}$, If $xy \leq 0$, then
 $x \leq 0$ or $y \leq 0$

Let $x, y \in \mathbb{R}$ $x \cdot y > 0$
Assume $x > 0$ and $y > 0$

$xy > 0y$
then $xy > 0$

Therefore, $\forall x, y \in \mathbb{R}$, If ... then

$\forall x, y \in \mathbb{R}$, If $x \leq y$ and $y \leq x$
then $x = y$

Let $x, y \in \mathbb{R}$

Assume $x \neq y$

we have $x < y$ or $y < x$

Therefore, If $x \leq y$ and $y \leq x$
 $x = y$

$\forall x \in \mathbb{R}$, If $\forall a \in (0, \infty)$, $x \leq a$
then $x \leq 0$

Let $x \in \mathbb{R}$

Assume $x > 0$

let $a = \frac{x}{2}$

Since $x > 0$, $0 < \frac{x}{2}$, $0 < a$

this proves $a \in (0, \infty)$

Since $x > 0 \rightarrow 2x > x$
 $x > \frac{x}{2}$

So $x > a$

Therefore If ... then

Existential Instantiation

$\exists x \in \mathbb{R}$, $P(x)$

Assume $\exists x \in \mathbb{R}, P(x)$

let $a \in \mathbb{R}$ for which $P(a)$

Ex.

$\forall x, y \in \mathbb{R}$, If $\exists a \in \mathbb{R}, x < a$ and $a < y$, then $x < y$

Let $x, y \in \mathbb{R}$

Assume $\exists a \in \mathbb{R}, x < a$ and $a < y$

Let $t \in \mathbb{R}, x < t$ and $t < y$

$x < t$ and $t < y \rightarrow x < y$

Therefore $x < y$

$\forall x, y \in \mathbb{R}$, If $0 < x$ and $\exists a \in (0, \infty)$

$\exists b \in (-\infty, a)$, $2ax < ay + bx$, then $x < y$

Let $x, y \in \mathbb{R}$

.... $x < y$

Assume $0 < x$, $\exists a \in (0, \infty)$

$\exists b \in (-\infty, \infty)$, $2ax < ay + bx$

Choose $s \in (0, \infty)$ and $t \in (-\infty, \infty)$

for which $2sx < sy + tx$

we have $0 < x$ and $0 < s$

and $t < s$ and $2sx < sy + tx$

Since $t < s$ and $0 < x$, we

have $tx < sx$ and $\underline{sy + tx < sy + sx}$

Now $2sx < sy + tx$ and $\downarrow$

$$2sx < sy + sx$$

$$sx < sy$$

$$s^{-1}sx < s^{-1}sy \rightarrow x < y$$

$\forall x \in \mathbb{R}, \text{ if } \exists a \in \mathbb{R}, 0 < a \leq x$
then $\exists b \in \mathbb{R}, 0 < b < x$

Let $x \in \mathbb{R}$

Assume $\exists a \in \mathbb{R}, 0 < a \leq x$

Choose $t \in \mathbb{R}, 0 < t \leq x$

Let $b = \frac{t}{2}$

Since $0 < t$, $0 + t < t + t$
 $t < 2t \quad \frac{t}{2} < t \rightarrow b < t$

Since $0 < t$, $0 < \frac{t}{2} \rightarrow 0 < b$

So, $0 < b < x$

Universal Instantiation

$\forall x \in \mathbb{R}, P(x)$

Assume $\forall x \in \mathbb{R}, P(x)$

Since $\square \in \mathbb{R}$, we have $P(\square)$

Ex

$\forall x \in \mathbb{R}$, If $\forall a \in \mathbb{R}, x \leq ax$
then $x = 0$

Let $x \in \mathbb{R}$

Assume $\forall a \in \mathbb{R}, x \leq ax$

Since $0 \in \mathbb{R}, x \leq 0 \cdot x$

Thus, $x \leq 0$

Since $2 \in \mathbb{R}, x \leq 2x$

$$x - x \leq 2x - x \quad 0 \leq x$$

So $x \leq 0$ and $0 \leq x$

$$x = 0$$

$\forall x, y \in \mathbb{R}$, If $\forall a \in (-\infty, x)$
 $x+a < y-1$, then $2x < y$

Let $x, y \in \mathbb{R}$

Assume $\forall a \in (-\infty, x), x+a < y-1$

Since $x-1 < x$, $x-1 \in (-\infty, x)$

Since $x-1 \in \mathbb{R}$, $x+(x-1) < y-1$

$$2x-1 < y-1$$

Then $2x < y$

$\forall x, y \in \mathbb{R}$, If $\forall a \in (-\infty, x)$
 $a < y$, then $\forall b \in (y, \infty), x < b$

Let $x, y \in \mathbb{R}$

Assume $\forall a \in (-\infty, x)$, $a < y$

Let $b \in (y, \infty)$

Then $y < b \rightarrow y+x < b+x$

So $y+x-b < x$

$y+x-b \in (-\infty, x)$

$y+x-b < y$ by Assumpt.

$x-b < 0 \rightarrow x < b$

Therefore

$\forall x \in \mathbb{R}$, If $\forall a \in (0, \infty)$, $0 \leq x+a$
then $0 \leq x$

Let $x \in \mathbb{R}$

Assume $\forall a \in (0, \infty)$, $0 \leq x+a$ and $x < 0$

$$\text{Since } x < 0 \quad \frac{x}{2} < 0 \rightarrow 0 < -\frac{x}{2}$$

$$\text{So, } -\frac{x}{2} \in (0, \infty)$$

$$0 \leq x + \left(-\frac{x}{2}\right) = 0 \leq \frac{x}{2}$$

$$0 \leq \frac{x}{2} \quad \text{and} \quad 0 < -\frac{x}{2}$$

contradiction

Archimedean Property

$$\forall a, b \in \mathbb{R}, \text{ If } 0 < a, \text{ then} \\ \exists n \in \mathbb{N}, b < na$$

$$\text{Let } a, b \in \mathbb{R}$$

$$\text{Assume } 0 < a \text{ and } \exists n \in \mathbb{N}, b \geq na$$

$$\text{Let } A = \{x \in \mathbb{R}, \exists n \in \mathbb{N}, x = na\}$$

$$\text{Let } B = \{x \in \mathbb{R}, \forall n \in \mathbb{N}, na \leq x\}$$

Let $x \in A$ and $y \in B$

Since $x \in A$, choose $m \in \mathbb{N}$, $x = am$

Since $y \in B$, $am \leq y$

Therefore $x \leq y$

$\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x < n$

Let $x \in \mathbb{R}$ $b < a \cdot n$

Since $0 < 1$, $x < 1 \cdot n$

Therefore $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x < n$

$\forall x \in \mathbb{R}$, If $0 < x$, then $\exists n \in \mathbb{N}$

$$\frac{1}{n} < x$$

Let $x \in \mathbb{R}$

Assume $0 < x$

Since $0 < x \rightarrow 1 < xn$

Let $n \in \mathbb{N}$ with $1 < xn$

$$\frac{1}{n} < x$$

Therefore...

$\forall x \in \mathbb{R}$, if $x < 1$, then $\exists n \in \mathbb{N}$

$$x < \frac{n}{n+1} < 1$$

Let $x \in \mathbb{R}$

Assume $x < 1$ then $0 < 1-x$

Archim. $x < (1-x)n$

Let $n \in \mathbb{N}$ with $x < (1-x)n$

$$x < n - xn$$

$$x + xn < n$$

$$x(1+n) < n \quad x < \frac{n}{1+n}$$

Direct

Assume P

Show Q

Therefore

Contraposition

Assume $\neg Q$

Show $\neg P$

Therefore

Contradiction

Ass. $P \wedge \neg Q$

Reach $\neg R$

Therefore

$\forall x, y \in \mathbb{R}$, If $3y \leq 2x$, then $x \leq 0$ or $y \leq x$

Contrapost

Let $x, y \in \mathbb{R}$... $3y > 2x$

Assume $x > 0$ and $y > x$

Since $x < y$ and $0 < 3$
 $3x < 3y$

Also $0 < 1$, so $2+0 < 1+2$

$2 < 3$, $0 < x \rightarrow 2x < 3x$

$2x < 3x < 3y$

Contrapos

$2x < 3y$

$\forall x, y \in \mathbb{R}$, If $2y \leq x+1$, then $x \leq 1$ or $y \leq x$

Let $x, y \in \mathbb{R}$

Assume $x > 1$ and $y > x$... $2y > x+1$

$$x < y$$

$$\underline{2x < 2y}$$

$$1 < x$$

$$x+1 < \underline{2x}$$

$$\underline{x+1 < 2y}$$

Contradiction

$\forall x, y \in \mathbb{R}$, If $\forall x \in \mathbb{R}$, $ax \neq 1$, then $a=0$

Let $x, y \in \mathbb{R}$

Assume $\exists x \in \mathbb{R}$, $ax \neq 1$ and $a \neq 0$

Since $a \neq 0$ $a^{-1} \in \mathbb{R}$

$$a \cdot a^{-1} = 1$$

and assume $a \cdot x \neq 1$

contradiction

Contradiction

$\forall x, y \in \mathbb{R}$, If $xy < x+y$ and $x < y$, then $x < 2$

Let $x, y \in \mathbb{R}$

Assume $xy < x+y$ and $x < y$ and $2 \leq x$

$$\begin{array}{ll} x < y & x+y < 2y \\ xy < x+y & \underline{xy < 2y} \end{array}$$

$$0 < 2 \leq x < y \quad 0 < y$$

Then $2 \leq x$, we $2y \leq xy$

$$xy < 2 \quad 2y \leq xy$$

contradiction

5. $\forall a \in \mathbb{R}, \exists b \in \mathbb{R}, b < a$

Let $a \in \mathbb{R}$

Assume $b = a - 1$

$$a-1 < a$$

$$-1 < 0$$

true

6.

$\forall a \in \mathbb{R}$, if $a < 1$, then $\exists b \in (0, \infty)$, $a+2b < 1$

Let $a \in \mathbb{R}$

Assume $a < 1$

$$\text{Let } b = \frac{1-a}{4}$$

7.

$\forall a \in \mathbb{R}$, if $a < 1$, then $\exists b \in \mathbb{R}$, $2a < b < 2$

Let $a \in \mathbb{R}$

Assume $a < 1$

$$\text{Let } b = a+1$$

Since $a < 1$, $a - a < 1 - a$
 $0 < 1 - a$

$$2a < a + 1 \quad \underline{2a < b}$$

Also $a < 1$, $a + 1 < 1 + 1$
 $a + 1 < 2$

$$\underline{b < 2}$$

$$2a < b < 2$$

8. $\forall \varepsilon \in (0, \infty), \exists \delta \in (0, \infty), \forall x \in \mathbb{R}, \text{ if } x - 2 < \delta$
then $3x - 6 < \varepsilon$

Let $\varepsilon \in (0, \infty)$

Assume $\delta = \frac{\varepsilon}{3}$

9. $\forall a \in \mathbb{R}, \text{ if } \forall x \in \mathbb{R}, ax = x, \text{ then } a = 1$

Let $a \in \mathbb{R}$

Assume $\forall x \in \mathbb{R}, ax = x$

Since $1 \in \mathbb{R}$, $a \cdot 1 = 1$

Therefore $a = 1$

10.

$\forall x \in \mathbb{R}$, if $\forall a \in \mathbb{R}$, $ax = a$, then

$$2x^2 + x = 2x + 1$$

Let $x \in \mathbb{R}$

Assume $\forall a \in \mathbb{R}$, $ax = a$

$$\underset{a}{x}(\underset{x}{2x} + \underset{x}{1}) = \underset{x}{2x} + \underset{x}{1}$$

Therefore ...

11.

$\forall a \in \mathbb{R}$, if $\exists x \in (0, \infty)$, $x < a$,
then $\forall x \in (-\infty, 0)$, $x + b \leq y$

Let $a \in \mathbb{R}$

Assume $\exists x \in (0, \infty)$, $x < a$

Let $b \in (0, \infty)$ with $b < a$
then $0 < b$

Let $x \in (-\infty, 0)$

$\forall x \in \mathbb{R}$, If $x < 1$, then $\exists n \in \mathbb{N}$,
 $x + \frac{2}{n} < 1$

Let $x \in \mathbb{R}$

Assume $x < 1$ $0 < 1 - x$

From Archimedean Prop

$$\begin{aligned}n+2 &< n \\2 &< n - xn \\2 &< (1-x)n\end{aligned}$$

$$\exists n \in \mathbb{N} \in \mathbb{N}, \textcircled{2} < (1-x)n$$

$\forall x \in \mathbb{R}$, If $2 < x$, then $\exists n \in \mathbb{N}$

$$2 < x - \frac{3}{n}$$

Let $x \in \mathbb{R}$

$$2n < xn - 3$$

$$3 < xn - 2n$$

$$3 < (x-2)n \quad \text{From A.P}$$

$$\exists n \in \mathbb{N}, 3 < (x-2)n$$

$$3 < xn - 2n$$

$$2n < xn - 3$$

$$2 < \frac{xn}{n} - \frac{3}{n}$$

$$2 < x - \frac{3}{n}$$

20.

$\forall x \in \mathbb{R}$, IF $1 < x$, then $\exists n \in \mathbb{N}, \frac{n+1}{n} < x$

Let $x \in \mathbb{R}$

Assume $1 < x$ $0 < x-1$

From Archimedean Prop.

$$\exists n \in \mathbb{N}, 1 < n(x-1)$$

$$n+1 < xn$$

$$1 < xn - n$$

$$1 < n(x-1)$$

$$1 < nx - n$$

$$n+1 < nx$$

$$\frac{n+1}{n} < x$$

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$\forall x \in \mathbb{R}$, If $3 < x$, then $\exists n \in \mathbb{N}$

$$\frac{3n+4}{n} < x$$

Let $x \in \mathbb{R}$

$$3n+4 < xn$$

$$4 < xn - 3n$$

Assume $3 < x$

$$4 < n(x-3)$$

From Arch. Prop

$\exists n \in \mathbb{N}$, $4 < n(x-3)$

Let $n \in \mathbb{N}$ with $4 < n(x-3)$

$$4 < nx - 3n$$

$$4 + 3n < nx$$

$$\frac{4+3n}{n} < x$$

$\forall x \in \mathbb{R}, \text{ if } x < 3, \text{ then } \exists n \in \mathbb{N},$

$$x < \frac{3n+5}{n+1}$$

$$3n + x < 3n + 5$$

$$x - 5 < 3n - 3n$$

$$x - 5 < n(3 - x)$$

Let $x \in \mathbb{R}$

Assume $x < 3$ $0 < 3 - x$

Let $n \in \mathbb{N}$ with $x - 5 < n(3 - x)$

15. $\forall x, y \in \mathbb{B}, \text{ if } y \Rightarrow \neg x, \text{ then}$

$$x \wedge \neg y = x$$

Let $x, y \in \mathbb{B}$

Assume $y \Rightarrow \neg x$

$$y \vee \neg x = \neg x$$

$$\neg y \wedge x = x$$

16. $\forall a, x, y \in B$, If $(\neg x) \wedge a \Rightarrow y$
then $(\neg y) \wedge a \Rightarrow x$

Let $a, x, y \in B$

Assume $\neg x \wedge a \Rightarrow y$

$$\neg y \Rightarrow x \vee \neg a$$

$$\neg y \wedge a \Rightarrow x \vee F$$

$$\neg y \wedge a \Rightarrow x$$

17. $\forall x, y, z \in B$, If $x \Rightarrow y$ and
 $\neg y \Rightarrow z$, then $x \vee \neg z \Rightarrow y$

Let $x, y, z \in B$

Assume $x \Rightarrow y$ and $\neg y \Rightarrow z$

$$\neg z \Rightarrow y$$

$$y \vee \neg z = y \vee y$$

$$y \vee \neg z \Rightarrow y$$

$$x \Rightarrow y$$

$$x \vee \neg z \Rightarrow y \vee \neg z$$

Proof by contradiction and the well ordering property.

$\forall x \in \mathbb{Z}$, If x is odd, then $\forall n \in \mathbb{N}$

x^n is odd

Let $x \in \mathbb{Z}$

Assume x is odd and x^n is not odd

Let $n \in \mathbb{N}$ be smallest with x^n is not odd

$x = 1$ is odd x^n not odd

x^{n-1} is odd

now x is odd, x^{n-1} is odd

let $a, b \in \mathbb{Z}$

$$x = 2a + 1$$

$$x^{n-1} = 2b + 1$$

$$\text{Let } c = bx + a$$

$$x^n = 2c + 1$$

23.

$\forall x \in \mathbb{R}$, if $2 < x$, then $\forall n \in \mathbb{N}$

$$2^n < x^n$$

Let $x \in \mathbb{R}$

Assume $2 < x$ and $\exists n \in \mathbb{N}$
 $x^n \leq 2^n$

